

Abhinand N

Lead Security Engineer

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Summary

Cybersecurity Specialist with 6+ years of experience in security review and testing of web, API, and mobile applications. Skilled in threat modeling and architecture assessments, complemented by leveraging LLMs to accelerate vulnerability analysis and streamline security workflows. Extensive expertise in mobile application security, including hardening, reverse engineering, and malware/stalkerware analysis. Delivered technical talks and hands-on workshops at Black Hat, SINCON, and CODE BLUE, with a strong track record of mentorship and community leadership. Built on a strong foundation in software development, with past open-source contributions through Google Summer of Code.

Work Experience

Traboda CyberLabs Pvt. Ltd.

Kochi, Kerala, India

⇒ Lead Security Engineer (2023-Present) | Senior Security Engineer (2021-23)

May 2021 - Present

- **AI-assisted Android application security analysis framework:** Led the internal development of a custom security analysis framework to strengthen the organization's Android application security review capabilities. Designed and built a hybrid analysis engine combining static and dynamic techniques to identify high-risk vulnerabilities at scale. Implemented a lightweight taint analysis engine with linear scanning and inter-procedural CFG analysis to detect insecure data flows from untrusted sources. Integrated AI and LLM-driven analysis, exploit generation, and dynamic validation to automate verification workflows, significantly improving detection accuracy and reducing manual review time across the AppSec pipeline.
- Built and embedded an **application security pipeline** into the SDLC and CI/CD workflow for three mobile applications, automating security gates at every stage. Integrated Semgrep for SAST on every commit, OWASP Dependency-Track for continuous SCA monitoring of third-party vulnerabilities, OWASP Noir to generate OpenAPI specifications from source for automated OWASP ZAP API security testing, MobSF for pre-release static analysis, and TruffleHog to scan for exposed secrets - ensuring continuous, shift-left security validation.
- **Vulnerability assessment and penetration testing for web, API, and mobile applications:** Conducted STRIDE-based threat modeling and security assessments of web applications (Node.js, Python/Django), APIs (REST, GraphQL), and mobile applications (Android/Java, iOS/Swift, React Native).
- Developed **novel forensic methodologies for extracting artefacts** from specific Android applications. Built AndroCop, an Android forensic acquisition tool that automates the collection of application data, databases, files, and other relevant forensic evidence from selected apps with a single-click workflow.
- **Redesigned CTF platform architecture** to enhance security, improve performance, and reduce infrastructure costs while preserving scalability and high availability. Hardened the platform by implementing secure configurations, access controls, and isolation mechanisms to prevent challenge manipulation and unauthorized access. **Automated CTF platform deployments** using Terraform and Ansible, integrating payment gateway support with secure transaction handling while ensuring scalable and reliable infrastructure provisioning.
- Designed and developed 10+ Web Security challenges covering OWASP Top 10 and 30+ customized mobile and API security challenges, covering topics such as reverse engineering, hardened root/jailbreak detection bypass, API exploitation, and vulnerability misconfigs covering MASVS.
- Contributed to the end-to-end delivery of CTF events for employee training, including challenge deployment, platform customization, and infrastructure automation for 20+ enterprises.

Amrita Center for Cyber Security Systems and Networks

Kollam, Kerala, India

⇒ Senior Security Researcher (2020-2021) | Research Assistant (2019-2020)

Dec 2019 - May 2021

- **Android Platform Security Lab:** Built and maintained an internal Android security testing environment by unlocking bootloaders, deploying custom recoveries and ROMs, and configuring root and hooking solution with Magisk, KernelSU, Zygisk, and LSPosed to support mobile application security testing, malware analysis, and platform security research.
- Co-authored the **IEEE paper Stalkerware Detection Using Static and Dynamic Analysis**, contributing to the research, implementation, and evaluation of static and dynamic techniques for stalkerware detection.
- Collaborative project with Avira by **analyzing and reporting 15+ malware applications** on the Google Play Store to the Coalition Against Stalkerware and Google Play Console.
- Designed course syllabus and content for a graduate-level **Mobile Security curriculum**.
- Designed and deployed hands-on web and mobile security assignments on a custom AutoLab instance, implementing automated grading workflows to enable scalable, consistent evaluation of student submissions.
- Regularly participated in CTF competitions, solving diverse mobile security challenges

NUS-Singtel Cyber Security Research Lab

Singapore

⇒ Intern

Aug - Oct 2019

- Built a DNS-over-HTTPS traffic classification pipeline: automated dataset collection from Alexa's top 10,000 sites with Selenium/Python, then cleaned, labelled, and engineered features from captured traffic.
- Analyzed a private Cisco ASA/PIX firewall log dataset, automating log summarization in Python/Pandas and surfacing malware indicators from anomalous connection patterns.

Trainings & Talks

2026	Trainer, BSides Kerala , Pins, Pads N Payloads: A Hands-on Hardware, IoT and OT Security	Kochi, India
2025	Trainer, CODEBLUE , Wired CTF: a combination of hardware, IoT and OT hacking arena	Japan
2025	Trainer, SINCON , Unlocking Android Apps: A Deep Dive into Hacking and Security	Singapore
2024	Speaker, bi0s-Meetup , Hunting Stalkerwares for Fun	Bangalore, India
2023	Speaker, BlackHat MEA , One-Click Forensic Investigation and Data Extraction Tool for Android	Saudi Arabia
2023	Speaker, BlackHat EU , One-Click Forensic Investigation and Data Extraction Tool for Android	United Kingdom
2016	Speaker, Fedora Users and Developers Conference , Contributing to Open Source and Git Fundamentals	Cambodia

Additional Experience

- **CVE-2025-20240**: Medium - Cross-site Scripting in Cisco
- **Organizer InCTF 2026** - India's first national-level Capture The Flag initiatives introducing 10,000+ students to practical cybersecurity through hands-on challenges, and bootcamps
- **Security Researcher at bi0s Pentest**: India's No.1 CTF team - vulnerability research and CTF participation (2020-Present)
- **Staff - Global Cybersecurity Camp 2025 (GCC)**: Represented India in a cybersecurity training program that brings together country-nominated cybersecurity professionals and students for intensive technical training and international collaboration.
- **MITRE ATT&CK**: Authored T1629.001 – Execution via Android Device Administrator, documenting Android adversary techniques for the ATT&CK knowledge base.
- **Red Team Certification**: Security Innovation, 2021
- **Summer School Participant**: Data Mining & Business Intelligence for Cyber Security Summer Program, Ben-Gurion University of the Negev, Israel.
- **Teaching Assistant**: 7th International Summer School on Information Security and Protection 2016

Open Source Activities

- 2018 **Selected team** for the Hackathon for Peace, Justice and Security, The Hague, Netherlands.
- 2016-17 **Google Code-In**: Mentored pre-university students competing in Wikimedia Foundation's open-source contest.
- 2016 **Google Summer of Code with Wikimedia Foundation**: Built and shipped EditNotify, an open-source MediaWiki extension that notifies users of wiki-page events
- 2016 **Organizer**: Wikimedia Hackathon Amrita University
- 2015-19 **FOSS@Amrita | amFOSS**: Active member of the open source club; mentored junior students contributing to open-source projects.

Education

B.Tech in Computer Science and Engineering | CGPA: 8.15

Amrita Vishwa Vidyapeetham

Kollam, Kerala, India

2015 - 2019

AISSE (Senior Secondary) | Score: 8.7

Bhavans Vidya Mandir, Chithali

Kerala, India

2015

Skills

Application Security	Semgrep, Burp Suite, OWASP ZAP, OWASP Noir, nmap, TruffleHog, gitleaks
Mobile Security	Reverse Engineering, code obfuscation, RASP, root/jailbreak detection, MobSF, JADX, Ghidra, Frida
DevSecOps	CI/CD Security Integration, Terraform, Ansible
Programming	Python, Java, JS, Bash
Cloud & Infrastructure	AWS, Docker, Kubernetes